

COMPOSITE MATERIAL DATASHEETS ADHESIVE EA9394 (LIQUID SHIM)

MATERIAL IDENTIFICATION

EU 1E001, 1E201 - This data is controlled under EU Dual Use Regulation 2021/821, as amended. Authorization may be required for the export of the data outside of the EU.

Short informal name	EA9394
Commercial name	Henkel LOCTITE EA9394 AERO
Specifications	AIMS10-07-001
Description	Two components paste epoxy-based adhesive, of curing temperature from RT to 90°C, for application as non-structural liquid shim
Version	0.1 (Jan/2025)

REFERENCES

- NT-B-ID-02015 Is. A "QUALIFICATION OF EA 9394 A/B (LIQUID SHIM) TO AIMS 10-07-001"
- SiRTAP screening tests program (G1c), not concluded or formally reported at release date of this datasheet

PHYSICAL PROPERTIES

Nominal cured thickness	0.1 mm	(*) Nominal 0.1mm, tolerance up to (0.1+0.2)mm = 0.3mm
Density	≤ 1400 kg/m ³	
Tg-onset (wet)	70 °C	(*) Function of curing temperature
Max. Operational Limit (MOL)	- °C	

STRESS/STRAIN CURVE OF ADHESIVE IN SHEAR

Shear modulus, stress and strain in linear limit, knee point and ultimate/failure points.

Property	Mean Values				B-basis K _B	E-KDF			B-basis Values			
	-55°C/Dry	RT/Dry	100°C/W	120°C/W		-55°C/Dry	100°C/W	120°C/W	-55°C/Dry	RT/Dry	100°C/W	120°C/W
G (MPa)	1100	980	280	100		1.122	0.286	0.102				
S _{lin} (MPa)	18.9	17.1	4.4	2.5		1.105	0.257	0.146				
S _{knee} (MPa)	43.6	37.0	9.4	6.6		1.178	0.254	0.178				
γ _{knee}	0.066	0.080	0.071	0.099	-	0.825	0.888	1.238		-	-	-
S _u (MPa)	65.6	40.7	14.8	10.8		1.612	0.364	0.265		-	-	-
γ _u	0.190	0.415	0.280	0.268	-	0.458	0.675	0.646		-	-	-

YOUNG MODULUS AND STRENGTH OF THE ADHESIVE IN TENSION & COMPRESSION

Property	Mean Values				B-basis K _B	E-KDF			B-basis Values			
	-55°C/Dry	RT/Dry	100°C/W	120°C/W		-55°C/Dry	80°C/W	120°C/W	-55°C/Dry	RT/Dry	100°C/W	120°C/W
E (MPa)												
F _{ty} (MPa)												
F _{tu} (MPa)												
E _c (MPa)	3900	2900		1300		1.345		0.448				
F _{cu} (MPa)	168	66.3		25.6		2.534		0.386				

SINGLE LAP SHEAR STRENGTH AND FRACTURE ENERGIES OF THE BONDED JOINT

Property	Mean Values				B-basis K _B	E-KDF			B-basis Values			
	-55°C/Dry	RT/Dry	100°C/W	120°C/W		-55°C/Dry	100°C/W	120°C/W	-55°C/Dry	RT/Dry	100°C/W	120°C/W
SLS (MPa)												
G _{1c} (N/mm)		1.0										
G _{2c} (N/mm)		1.0										

Notes:

- Some Mechanical properties improved in most recent re-qualifications of this material
- The G_{1c} value of 1000 J/m² is selected from SiRTAP screening tests program as representative value of the configurations tested. The G_{2c} value of 1000 J/m² is not an experimental data, but an expected lower bound , this fracture energy should not be lower than the G_{1c} value from observations in other epoxy based resins used in composites.

FM-73 @ 180°F

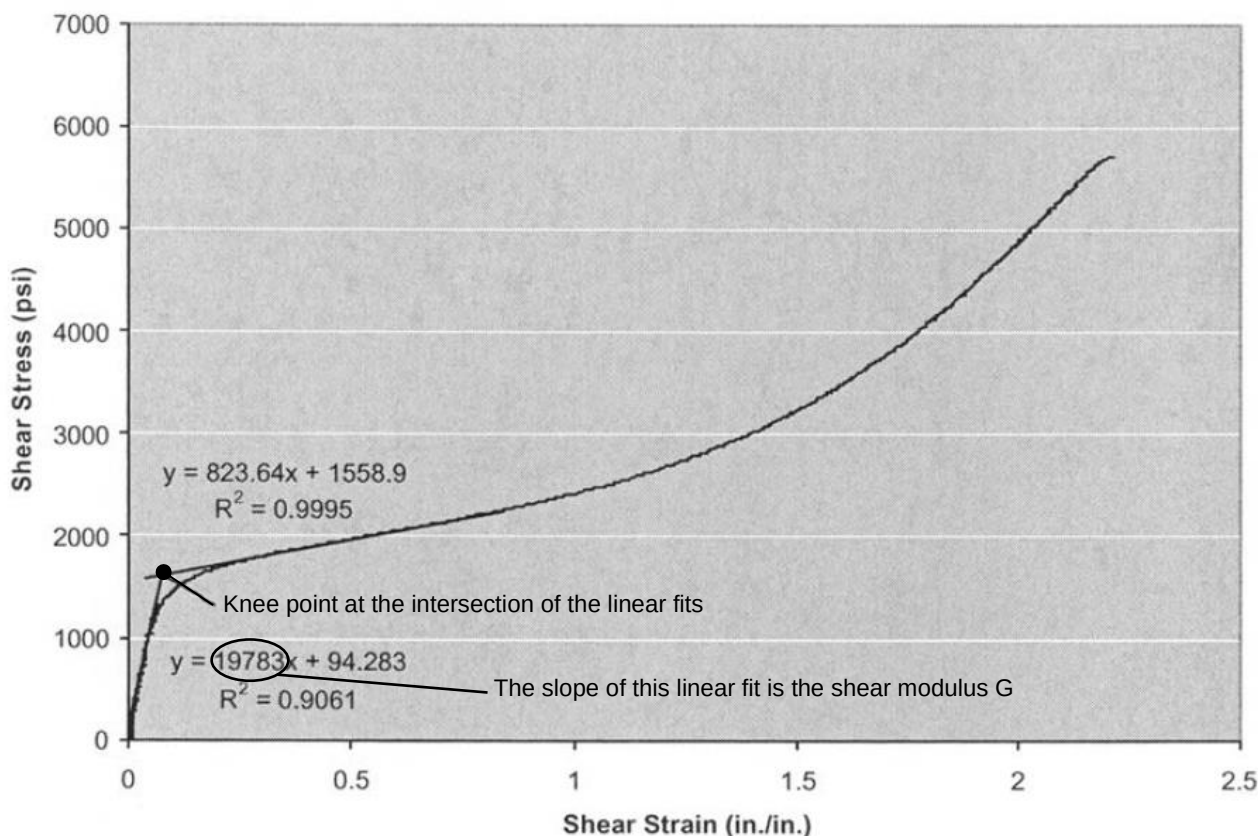


Figure 1. Example of shear stress-strain curve as defined per ASTM D5656 test method